
An Analysis of Federated Learning Techniques

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Abstract

Federated learning enables collaborative model training across decentralized, privacy-sensitive data silos without centralizing raw data. In this report, we extensively evaluate FL techniques; we first establish the empirical equivalence between FedSGD and centralized SGD; however, FedSGD remains impractical in realistic settings due to its communication overhead, motivating more communication-efficient alternatives. We establish FedAvg as our baseline and analyze its convergence under varying heterogeneity. By generating non-IID client splits through Dirichlet sampling, we characterize how increasing skew amplifies drift and affects overall performance. We then study extensions of FedAvg aimed at mitigating data heterogeneity. For FedProx, we find accuracy and convergence behavior to be nearly indistinguishable from FedAvg across a broad range of μ , while highlighting that large μ reduces client drift at the cost of accuracy. For FedSAM, sharpness-aware local training consistently improves accuracy. SCAFFOLD delivers the strongest improvements by explicitly correcting client drift, while FedGH provides intermediate gains by resolving conflicting client updates, outperforming FedProx but remaining below SCAFFOLD and FedSAM in our evaluations. Overall, accuracy follows a consistent trend across heterogeneous settings: SCAFFOLD > FedSAM > FedGH > FedProx $\approx$ FedAvg. Code and reproducibility material are available at: [code](#).

1. Introduction

Federated learning (FL) enables multiple clients to collaboratively train models without centralizing raw data. It faces three main challenges: (i) communication efficiency;

(ii) statistical heterogeneity, where non-identical client distributions induce client drift and impair convergence and generalization; and (iii) privacy leakage.

In this report, we study the effect of statistical heterogeneity and how different FL algorithms aim to target it. We first revisit a fundamental baseline by comparing FedSGD to centralized SGD, showing that under identical initialization and hyperparameters their optimization trajectories are empirically indistinguishable, yet FedSGD is communication-inefficient in practice. Beyond FedSGD, we adopt FedAvg as our primary communication-efficient baseline and use it to characterize how heterogeneity influences convergence. By generating controlled non-IID client splits through Dirichlet sampling, we directly observe how increasing skew amplifies client drift and slows optimization, providing a reference point for all subsequent methods.

We then evaluate methods that explicitly target heterogeneity. FedProx introduces a proximal term that penalizes deviations from the server model. In our setting, it behaves nearly identically to FedAvg for small μ and exhibits the expected trade-off at larger values, where drift is reduced at the cost of suppressed accuracy. FedGH applies gradient harmonization to remove pairwise conflicting client updates before aggregation improving upon FedAvg under strong non-IID splits while incurring no additional communication, but remains less effective than algorithms with explicit drift correction. SCAFFOLD, which uses control variates to correct client-specific update bias, provides the strongest drift mitigation and the largest accuracy gains across heterogeneous regimes, but with increased communication. Finally, FedSAM (Qu et al., 2022), which applies sharpness-aware local training, consistently improves accuracy, particularly under non, IID data, while incurring increased drift and higher local computation.

2. Methodology

2.1. FedSGD vs. Centralized SGD

We evaluate whether FedSGD with full client participation and a single local step per round ($K=1$) is equivalent to a centralized model trained with SGD on pooled data. To test this, we run two identical training loops with the same model, initialization, optimizer.

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We use a 3-layer NN classifier trained on MNIST with IID client partitions. Both setups use a fixed learning rate $\eta = 0.01$ and $T=15$ update rounds. In FedSGD, the training set is partitioned across $M=6$ clients; at each round t , the server broadcasts θ_t to all clients, each client computes a full-batch gradient on its local data, the server forms a weighted average of the returned gradients, updates the global parameters, and then broadcasts the updated parameters. In the centralized baseline, a single model is trained on the combined dataset. To test for equivalence, after each round we record (i) training loss, (ii) test accuracy on the MNIST test set, and (iii) the ℓ_2 norm of the parameter vector $\|\theta_t\|_2$. These metrics provide a quantitative measure of whether the two procedures follow near-identical optimization paths under matched conditions.

2.2. Implementing FedAvg

We implemented the Federated Averaging (FedAvg) algorithm to train the same 3-layer NN on MNIST using a local SGD optimizer with learning rate $\eta = 0.01$ and cross-entropy loss. We evaluated communication reduction strategies over $T = 30$ rounds under two data distributions using a 24,000-sample subset: (1) an IID partition across $M = 3$ clients and (2) a Non-IID partition (Dirichlet, $\alpha = 0.1$) across $M = 5$ clients. On the IID set, we first varied local epochs $K \in \{1, 5, 10, 20\}$ with full client participation ($f = 1.0$). We then analyzed partial client sampling $f \in \{1.0, 0.5, 0.2\}$ (with $K = 5$) on both the IID and Non-IID sets.

2.3. Exploring Data Heterogeneity

We use the same 3-layer neural network architecture for all experiments in this experiment. We evaluate all methods with the same initialization and training schedule as described in Section 2.2 with value of $K = 5$.

To analyze the effect of data heterogeneity on FedAvg, we train the same model five times using Dirichlet-based client partitions. For each run, class-probability vectors are sampled from a Dirichlet distribution with concentration parameter α , which determines the class-proportion allocation across clients. We consider $\alpha \in \{0.1, 0.3, 0.5, 1, 10\}$, where smaller values induce highly non-IID splits and larger values approach IID behavior.

For each α , we measure (i) the test accuracy of the global model per communication round and (ii) the client drift, computed as the mean L2 distance between each client's locally updated parameters and the global model parameters at the start of that round.

2.4. Mitigating Heterogeneity — FedAvg Extensions

We also extensively evaluate extensions of FedAvg that explicitly target reduction in drift through regularizers or control variates or aim to improve accuracy through sharpness aware training. Specifically, we evaluate: (i) FedProx (ii) SCAFFOLD (iii) FedGH (iv) FedSAM. To ensure consistency, all methods are evaluated on MNIST using the same 3 layer NN classifier, with identical initialization and training schedules as in the baseline (seed set to allow for reproducibility). We fix the local steps to $K = 5$ for all training regimes with $M=5$ clients and a learning rate of $\eta = 0.01$ across $T = 30$ training rounds. We consider two partitions: (i) IID splits across clients and (ii) non-IID splits through Dirichlet sampling with $\alpha=0.1$. We evaluate these methods against a FedAvg baseline trained under identical conditions, on a per communication round basis. We record (i) test accuracy and (ii) client-drift.

2.4.1. FEDPROX

We evaluate whether the proximal regularization in FedProx mitigates client drift and improves test accuracy relative to FedAvg under identical conditions. FedProx modifies each client's loss with a proximal term that penalizes drifts from the global model. At round t with server weights θ_t , client i optimizes

$$\min_{\theta} f_i(\theta) + \frac{\mu}{2} \|\theta - \theta_t\|_2^2, \quad (1)$$

where f_i is the local loss and μ controls the strength of the proximal regularization. We evaluate FedProx across $\mu \in \{0.001, 0.01, 0.1, 0.5\}$ in both IID and non-IID conditions.

2.4.2. SCAFFOLD

We also implement the SCAFFOLD algorithm to mitigate client drift. The server and clients initialized local (c_i) and global (c) control variates. During local training, client gradients were corrected using these variates ($g_i \leftarrow g_i + c - c_i$). After training, clients updated their c_i based on their model delta, and the server updated c by averaging the new c_i vectors. This was done for the partitions.

2.4.3. FEDGH

We evaluated whether FedGH can be used to mitigate client drift. In order to implement Gradient Harmonization, after each communication round, we collected client updates in the form of flattened gradient vectors. For every unordered client pair (i, j) , we checked whether the dot product of their gradients was negative, indicating a conflict. In such cases, we projected each gradient to remove the conflicting component by applying symmetric corrections:

$$g_i \leftarrow g_i - \frac{\langle g_i, g_j \rangle}{\|g_j\|^2} g_j, \quad g_j \leftarrow g_j - \frac{\langle g_i, g_j \rangle}{\|g_i\|^2} g_i.$$

This updated gradient vector was used to do further calculations.

2.4.4. FEDSAM

Finally, we evaluate FedSAM a method that tries to improve generalization by incorporating sharpness-aware training at a per client level with the assumption of shared global flatness. FedSAM applies the SAM objective on each client, seeking parameters robust to worst-case perturbations within a radius ρ . Each client optimizes the following objective:

$$\min_{\theta} \max_{\|\epsilon\|_2 \leq \rho} f_i(\theta + \epsilon), \quad (2)$$

We use a first-order approximation for ϵ . FedSAM aggregates the gradients/parameters exactly as FedAvg. We test our setup for $\rho \in \{0.01, 0.05, 0.1\}$ in both partition regimes. These values were chosen since they are common in SAM and FL literature (Foret et al., 2021).

3. Results

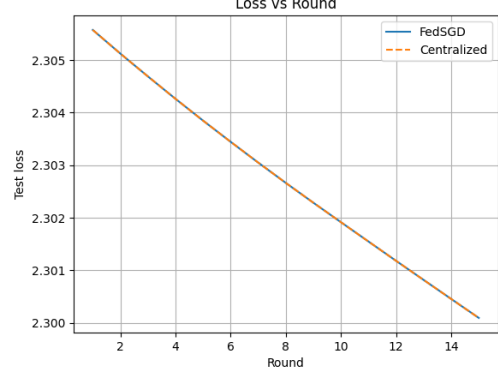
3.1. FedSGD vs. Centralized SGD

Under identical conditions, FedSGD and centralized SGD are identical. As shown in Figure 7, the test-loss and test-accuracy trajectories overlap. Concretely, test loss decreases from 2.3056 to 2.3001 over $T=15$ rounds for both methods, while test accuracy increases from 0.0722 to 0.1065. The parameter gap remains numerically negligible throughout - ranges from 1×10^{-6} to 1×10^{-8} . These minute differences are due to floating-point rounding and such issues. Therefore, we validate that centralized SGD and FedSGD are empirically equivalent.

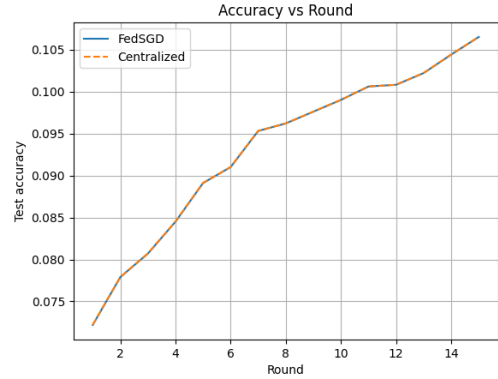
3.2. FedAvg

Our experiments illustrate the trade-offs among local computation, client sampling, and data distribution. On the IID MNIST task ($N = 3$), increasing the number of local epochs K from 1 to 20 improved both final test accuracy and convergence speed. Specifically, test accuracy rose from 90.35% ($K = 1$) to 94.50% ($K = 5$), 96.02% ($K = 10$), and 96.81% ($K = 20$). To reach approximately 90% test accuracy, $K = 1$ required 30 rounds, $K = 5$ required 5 rounds, $K = 10$ required 3 rounds, and $K = 20$ required 2 rounds. These gains came at the cost of increased client drift, with the average L2-norm drift in the first round increasing from 0.14 ($K = 1$) to 0.9670 ($K = 5$), 2.0279 ($K = 10$), and 2.82 ($K = 20$).

Fixing $K = 5$ on the same IID dataset, partial client sampling significantly improved communication efficiency. Full participation ($f = 1.0$) achieved 94.50% accuracy over 90 total client communications, whereas $f = 0.5$ (60 communications) and $f = 0.2$ (30 communications) achieved



(a) Test loss vs. round



(b) Test accuracy vs. round

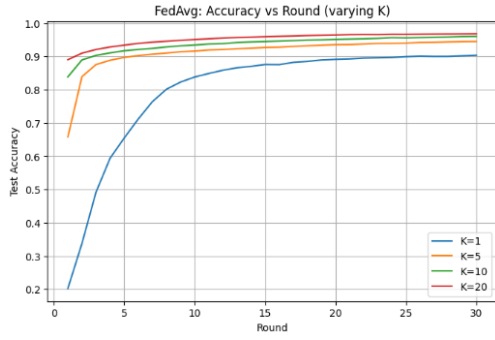
Figure 1. FedSGD vs. centralized SGD: loss and accuracy.

nearly identical final accuracies of 94.37% and 94.39%, respectively, with comparable final drift.

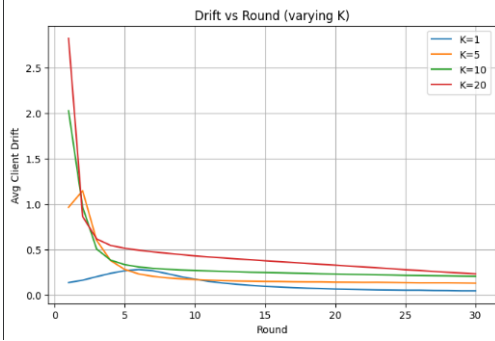
In contrast, on a highly Non-IID partition ($N = 5, \alpha = 0.1$), partial participation markedly reduced performance. Full participation ($f = 1.0$, 150 communications) achieved 92.76% accuracy with a final drift of 0.26. Halving participation ($f = 0.5$, 90 communications) decreased accuracy to 88.19% and increased final drift to 0.33, while minimal participation ($f = 0.2$, 30 communications) resulted in poor convergence, ending at 75.97% accuracy with a final drift of 0.37. These results indicate that while partial participation is effective for IID data, it introduces instability and slows convergence under heterogeneous data distributions.

3.3. Exploring Data Heterogeneity

Fig 4 visualizes accuracies against round number for our 5 models trained with different values of α . A clear trend is visible in our graph: Models that were trained with lower values of α , therefore more data heterogeneity, perform worse than models that were trained with higher values of α . For instance, the model trained with $\alpha = 0.1$ achieves a test accuracy of 0.6, whereas the model trained with $\alpha = 10$



(a) Accuracy vs. Round



(b) Avg Client Drift vs. Round

Figure 2. FedAvg Metrics with varying K.

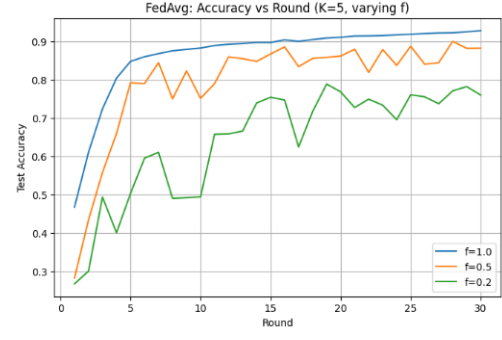
Method	Accuracy
FedAvg (baseline)	0.8736
FedProx ($\mu = 0.001$)	0.8736
SCAFFOLD	0.9148
FedGH	0.8845
FedSAM ($\rho = 0.1$)	0.8980

 Table 1. Final (highest) test accuracy across algorithms in non-IID ($\alpha = 0.1$) setting.

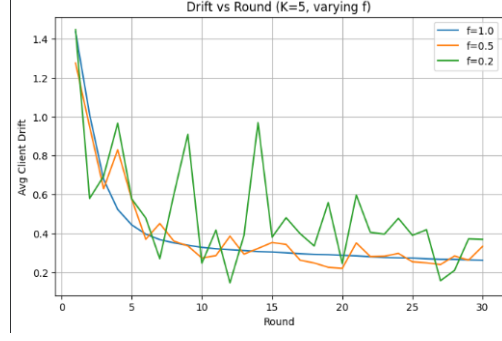
achieves an accuracy of 0.81. The inverse relationship holds true for Fig 5 which shows client drift against rounds for different values of α . Models trained with lower values of α exhibit a higher client drift.

3.4. Mitigating Heterogeneity — FedAvg Extensions

Under a common configuration of $M=5$ clients, $K=5$ local steps, and $T=30$ communication rounds, the FedAvg baseline attains 0.8736 accuracy on the non-IID split (Dirichlet $\alpha=0.1$) and 0.9296 on the IID split. The combined results for the non-IID split across methods are reported in Table 1, and detailed per-method results are presented in the subsections that follow.



(a) Accuracy vs. Round



(b) Avg Client Drift vs. Round

Figure 3. FedAvg Metrics with varying f.

3.4.1. FEDPROX

Across both partitions, FedProx behaves very similar to FedAvg in accuracy while reducing drift as μ increases, revealing a trade-off (Table 2). In the non-IID case, final accuracies remain effectively equal to FedAvg’s baseline of 0.8736 for small $\mu \in [0.001, 0.01]$. Larger μ introduces mild accuracy loss ($\mu=0.1$: 0.8724; $\mu=0.5$: 0.8683 but drift decreases with μ : (−0.4%) to (−3.6%), and (−14.5%) respectively. For the IID case, a similar trend is followed - accuracies are essentially unchanged for small μ (0.9296, with a slight drop for larger values ($\mu=0.1$: 0.9291; $\mu=0.5$: 0.9264. Drift reductions are more pronounced as μ grows with a reduction of (−17.5%) at $\mu=0.5$. In short, moderate $\mu \in [0.01, 0.1]$ preserves FedAvg-level accuracy while shaving 3–4% drift; aggressive $\mu=0.5$ yields larger drift reductions (≈ 15 –18%) at a small accuracy drop.

Method	Non-IID ($\alpha=0.1$)	IID
FedAvg	0.8736 / 0.2279	0.9296 / 0.1133
$\mu=0.001$	0.8736 / 0.2278	0.9296 / 0.1133
$\mu=0.01$	0.8736 / 0.2270	0.9296 / 0.1129
$\mu=0.1$	0.8724 / 0.2196	0.9291 / 0.1089
$\mu=0.5$	0.8683 / 0.1948	0.9264 / 0.0935

Table 2. FedProx final accuracy / avg drift.

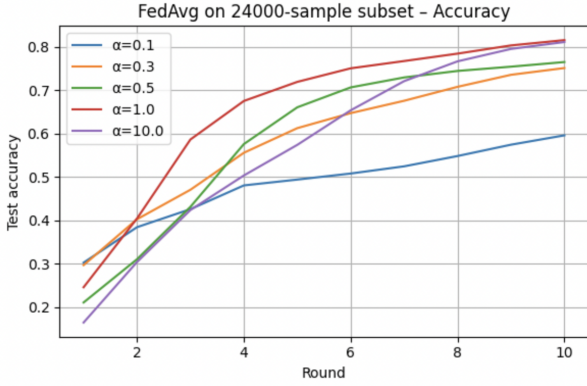


Figure 4. Accuracy per round (Dirichlet-based heterogeneity setup).

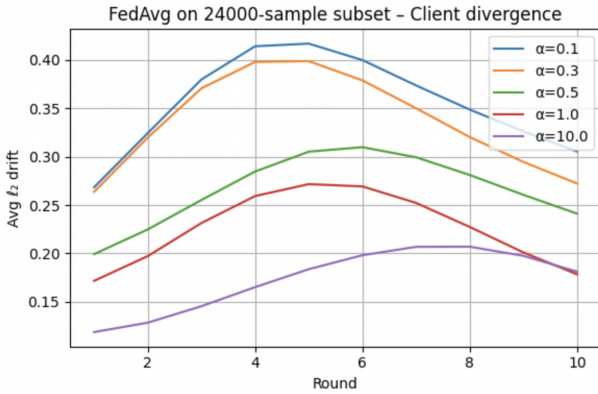


Figure 5. Client Drift per round (Dirichlet-based heterogeneity setup).

3.4.2. SCAFFOLD

SCAFFOLD performed the best across all algorithms and outperformed both FedAvg and FedProx, demonstrating its effectiveness in mitigating client drift. It achieved the highest final test accuracy at 91.48%. The accuracy plot shows SCAFFOLD’s performance curve diverging from the others around round 8, indicating faster convergence to a higher ceiling. The drift plot shows the most dramatic difference: while all methods began with a high average L2-norm drift (≈ 1.0), SCAFFOLD’s drift-correction mechanism resulted in a rapid decrease to a final value of just 0.0685. In contrast, FedAvg’s drift plateaued at 0.2279. While all methods used 150 total client updates, SCAFFOLD incurred a 2x communication overhead (131.3M vs. 65.6M bytes) due to the transmission of control variates (c_i). In summary, for this highly heterogeneous setting, SCAFFOLD’s 2x communication cost was justified by a significant +4.12% gain in final accuracy and a 3.3x reduction in final client drift compared to FedAvg.

3.4.3. FEDGH

FedGH produces consistent accuracy gains over FedAvg while offering modest drift reduction. Under the non-IID partition ($\alpha=0.1$), FedGH reaches a final accuracy of 0.8845, improving upon FedAvg’s 0.8736. Drift is also lower: 0.2049 compared to FedAvg’s 0.2279, indicating that gradient harmonization effectively suppresses conflicting client updates without materially slowing convergence. The same trend holds in the IID case: the final accuracy is 0.9320 versus FedAvg’s 0.9296, and the final drift remains close (0.1142 vs. 0.1133). Compared with other algorithms listed in Table 1, FedGH outperforms FedProx but loses to both FedSam and SCAFFOLD.

3.4.4. FEDSAM

FedSAM improves accuracy relative to FedAvg, especially in the heterogeneous case (Fig. 6), achieving the second highest accuracy overall (right after SCAFFOLD)). For the non-IID case, accuracy increases with ρ : 0.8788 at $\rho=0.01$, 0.8899 at $\rho=0.05$, and 0.8980 at $\rho=0.1$ over the baseline FedAvg. Drift also rises with ρ : 0.2279 in FedAvg to 0.3220 (+41.3%) in the $\rho = 0.5$ case. In the IID setting, gains are smaller but still consistent: from 0.9349 (FedAvg) to 0.9355 ($\rho=0.01$), 0.9372 ($\rho=0.05$), and 0.9372 ($\rho=0.1$). Drift increases mildly but FedSAM offers accuracy improvements under both IID and non-IID partitions, with the strongest gains under non-IID data, but at the cost of increased drift and training times.

4. Discussion

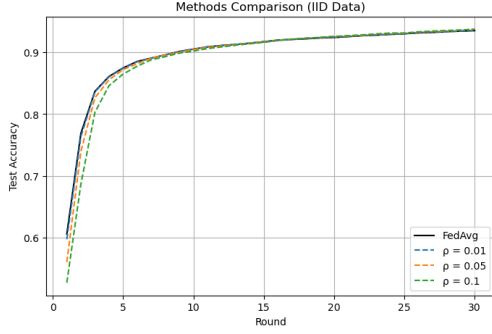
4.1. FedSGD vs. Centralized SGD.

We show that FedSGD and Centralized SGD are empirically equivalent too - the only differences are caused by factors like floating-point rounding and gradient accumulation. Practically, however, FedSGD is communication-inefficient because it requires a synchronization every round, without reducing communication overhead through any local steps or compression (Kairouz et al., 2021; ?). This calls for more communication efficient methods that handle client drift as well.

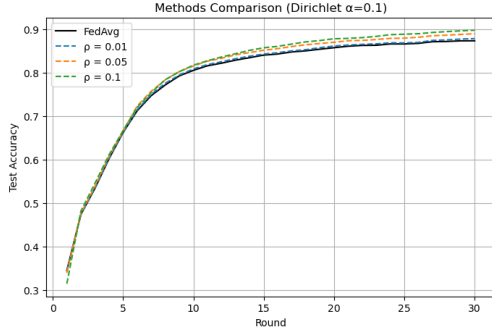
4.2. Impact of data heterogeneity on FedAvg.

Moving from IID to Dirichlet non-IID with $\alpha=0.1$ drops the final FedAvg accuracy and increases the average drift. This is the classic “client drift” phenomenon: client objectives differ, so local updates point in conflicting directions, resulting in increased variance. The result in a lower eventual test accuracy, which we also show through our experiments (Zhao et al., 2018b).

Our analysis of FedAvg highlights two primary trade-offs:



(a) IID partition



(b) Non-IID partition

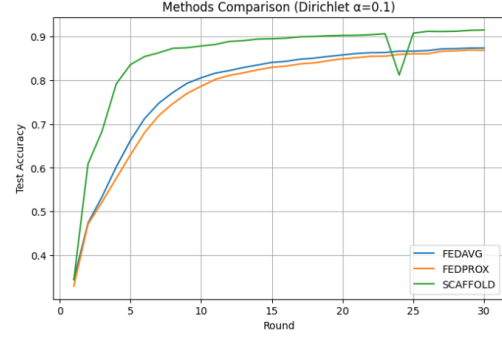
Figure 6. FedSAM test accuracy in IID and non-IID case

one between local work (K) and client drift, and another between communication efficiency (f) and stability, which is critically dependent on the data distribution.

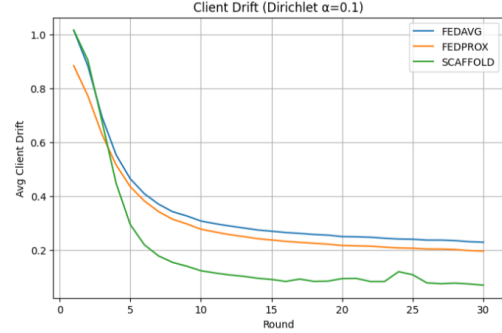
Increasing local epochs (K) on IID data presents a favorable trade-off. As seen in the "Accuracy vs Round (varying K)" plot, increasing K from 1 to 20 accelerates convergence (in rounds) and improves final accuracy from 90.35% to 96.81%. The cost is significantly higher client drift (first-round drift grew from 0.14 to 2.82). This works because, with IID data, clients are unbiased estimators; even though they diverge locally, their average converges to a superior global model.

In contrast, client sampling (f) reveals a trade-off between communication cost and model stability that is governed by data distribution. On the IID dataset ($N = 3$), this trade-off is highly advantageous. Reducing participation from $f = 1.0$ (90 total comms) to $f = 0.2$ (30 comms) cut communication by 3x with no meaningful drop in accuracy (94.50% vs. 94.39%). This is because any random client is a good proxy for the global objective.

However, this strategy fails on the highly Non-IID partition ($N = 5, \alpha = 0.1$). Here, reducing participation from $f = 1.0$ (150 comms) to $f = 0.2$ (30 comms) was disastrous, dropping accuracy from 92.76% to 75.97% and increasing final drift from 0.26 to 0.37. The reason is that each client update is *biased*. Full participation is essential to average out



(a) Accuracy vs. Round across different methods



(b) Avg Client Drift vs. Round across different methods

Figure 7. A comparison of FedAvg, FedProx and Scaffold over accuracy and client drift

these competing biases. With partial sampling, as seen in the "Accuracy vs Round (varying f , Non-IID)" plot, the global model is pulled by the skewed objectives of the sampled subset, leading to instability and poor convergence.

In conclusion, increasing K is generally effective for reducing rounds, but client sampling ($f < 1.0$) is only a viable communication-saving strategy on IID data.

4.3. Exploring Data Heterogeneity

Our results are backed by existing literature. Works like (Zhao et al., 2018a) show that under non-IID settings, accuracy can drop by as much as 55%, which is consistent with our observed 22% decrease. They explain this drop through an increase in weight divergence between local and global models when data heterogeneity is high. We observed the same trend, as α decreases, client drift increases, implying higher weight divergence, which explains the reduction in accuracy.

This behavior can be understood by looking at how local gradients behave under heterogeneous data. When clients have very different data distributions, each client's gradient vector points in a different direction, toward its own local minimum. As a result, the aggregated gradient vector becomes misaligned with the global gradient direction, leading

to poor convergence and lower overall accuracy. The same gradient misalignment also accounts for the rise in client drift, since each local model moves away from the global model in a different direction during local updates.

Moreover, some works exhibit unstable and slow convergence under heterogeneous data (Li et al., 2020b). However, these studies typically use a large number of local epochs and a large number of participating clients. The fact that our models did not have unstable convergence can be explained by our small values of K and M , 5 and 3, respectively.

4.4. Mitigating Heterogeneity — FedAvg Extensions

FedProx adds a regularization through a proximal term that penalizes large local updates (Li et al., 2020a). In our runs, FedProx tracks FedAvg’s accuracy almost exactly for small $\mu \in \{0.001, 0.01\}$, while delivering modest drift reductions (up to $\sim 4\%$ relative). Larger μ strengthens the penalty, which reduces the drift albeit at the cost of a small accuracy drop. These trends match FedProx’s convergence analysis, where the proximal term controls gradient dissimilarity at the cost of shrinking effective step size. Moreover, we observed stable training without overshooting and a convergence similar to the FedAvg baseline. This makes sense since we used full participation and literature suggests that partial participation, larger K or stronger heterogeneity can worsen drift, stability and convergence (Kairouz et al., 2021). It is important to note that the similar trajectories can also be associated with the choice of dataset - MNIST is a relatively *easier* dataset because of which the gains of FedProx were not as apparent. However, this also highlights the importance of the dataset and choosing the correct μ such that it is not identical to FedAvg or makes local training effective (Li et al., 2020a). The effectiveness of FedAvg lies in its simplicity and little to no overhead - it is nearly free computationally: it reuses FedAvg’s local loop and adds only a proximal term without affecting communication. This increases its practicality makes it extremely practical but results in marginal gains as highlighted by our experiments.

Our comparative analysis on the highly Non-IID dataset ($\alpha = 0.1$) highlights how each algorithm manages client drift and the resulting trade-offs with the inclusion of Scaffold. It directly addresses this bias using global (c) and local (c_i) control variates to estimate and correct drift through the adjusted update $g_i \leftarrow g_i + c - c_i$. This explicit correction sharply reduces drift to 0.07 and drives accuracy up to 91.48%. SCAFFOLD, while doubling communication overhead (131.3M bytes) to transmit control variates, achieves a +4.12% accuracy gain and $3.3\times$ drift reduction, yielding faster convergence and substantially improved robustness. In this highly Non-IID setting, the added cost is clearly justified by its superior performance.

FedGH offers a middle ground between the light regular-

ization of FedProx and the stronger correction mechanisms of SCAFFOLD and FedSAM. Its gradient-harmonization step removes pairwise conflicting components from client updates, which suppresses oscillation in heterogeneous settings without adding any auxiliary variables or extra communication. This explains why it consistently outperforms FedProx under non-IID data: whereas the proximal term in FedProx only restricts the magnitude of local drift, FedGH directly reduces gradient disagreement, yielding a more aligned global update and therefore better robustness to skewed client distributions.

However, FedGH remains weaker than SCAFFOLD and FedSAM because it only resolves conflict at the level of client-client interactions, not at the level of global bias. SCAFFOLD corrects the drift explicitly via control variates and therefore eliminates systematic update bias accumulated over rounds—something FedGH cannot address. Likewise, FedSAM’s sharpness-aware perturbation improves generalization by biasing updates toward flatter regions of the loss surface, which provides an accuracy gain that gradient harmonization alone does not deliver.

Lastly, FedSAM improves final accuracy in both regimes, with larger gains under heterogeneity, which is consistent with literature suggesting that it provides an improvement on FedAvg baselines (Qu et al., 2022). For FedSAM, we also observe stable training with similar convergence, which can be associated to the same aforementioned reasons. Interestingly, we observed diminishing returns in the IID case with increasing values of ρ and marginal gains in accuracy. However, the downside of this method is that it requires more compute and training time as the gradients are computed twice, which can make it impractical for constrained devices. Furthermore, it relies on the assumption of a global shared flat minima, which might not be true for more challenging, skewed data and thus, similar gains might not be observed. Lastly, as suggested by the original paper, different datasets perform optimally under different values of ρ , which further highlight the need for careful tuning of hyper-parameters, which if not done correctly can result in sub-optimal performance (Qu et al., 2022).

5. Conclusion

We evaluated several FL algorithms under matched settings to isolate the effect of statistical heterogeneity. FedSGD was shown to be equivalent to centralized SGD in optimization behavior, but impractical due to its communication cost. FedAvg served as a strong baseline, with Dirichlet-based experiments confirming that increasing heterogeneity amplifies client drift and slows convergence.

Among the extensions, FedProx offered only minor improvements over FedAvg. FedGH provided moderate gains

by reducing conflicting client updates, improving robustness under non-IID data while maintaining FedAvg-level communication. FedSAM consistently improved accuracy, especially in heterogeneous regimes, though at the cost of higher drift and computation. SCAFFOLD delivered the strongest drift correction and the highest final accuracy, but with increased communication overhead.

In summary, the methods follow a clear trend under non-IID data: SCAFFOLD > FedSAM > FedGH > FedProx $\approx$ FedAvg, reflecting the trade-offs between communication, computation, and heterogeneity robustness.

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